

FOR MORE WORKBOOKS, (Librarian) 0703181364
SCIENCE SCHOLARS ACADEMY
PERIOD-3 ELEMENTS:
(Lesson 2)

Lesson targets

(1). Reaction of elements with $\rightarrow$ H_2O ✓
 $\rightarrow$ dilute Acids ✓
 $\rightarrow$ Alkalies *

(*) Resolving concepts like;
 $\rightarrow$ disproportionation rxn ✓
 $\rightarrow$ oxidation states ✓

(2). Reaction of oxides $\rightarrow$ H_2O *
 $\rightarrow$ dilute Acids *
 $\rightarrow$ Alkalies *

(*) Hint MORE on preparation! *

(3) Research, next stop & wrap-up! ✓

END

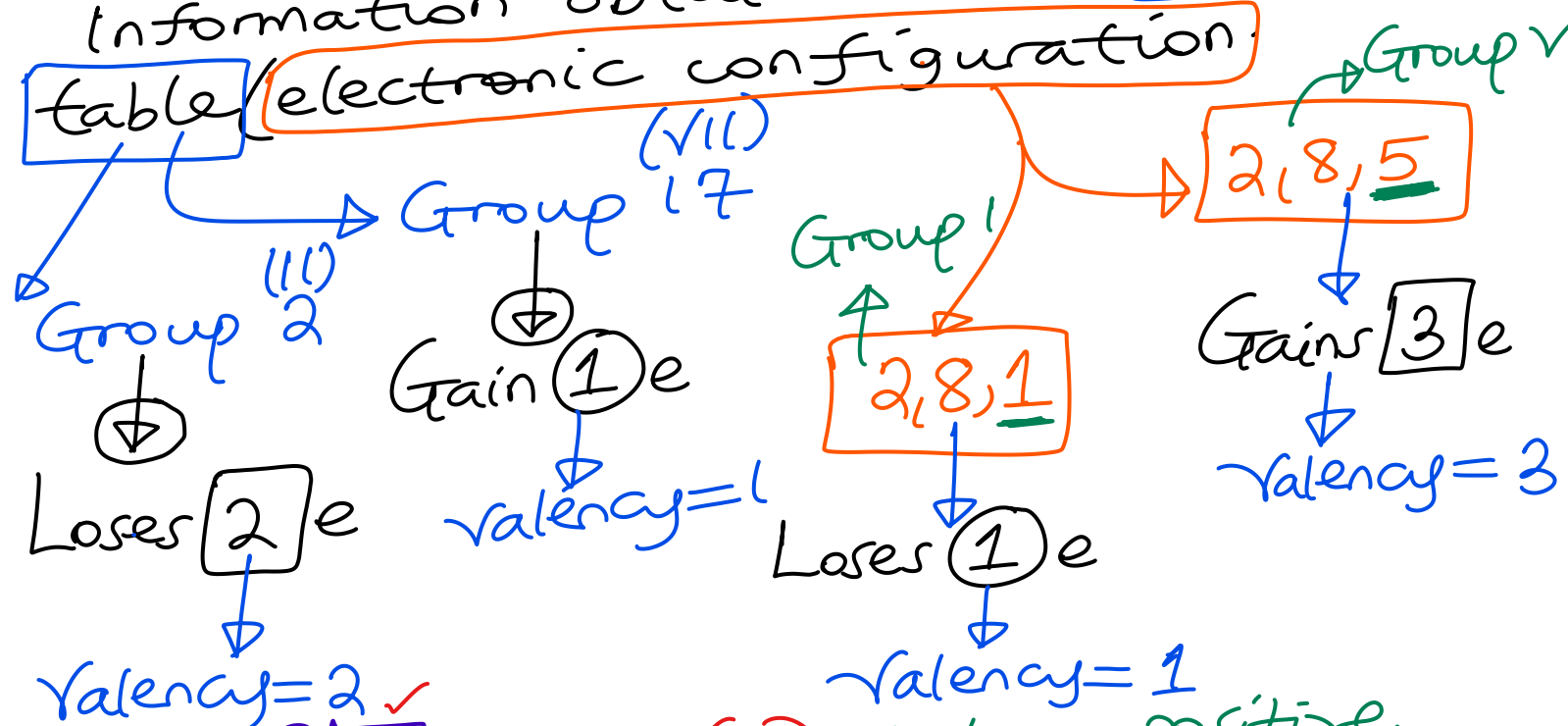
we are **Locally** learning them
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OXIDATION STATES: today!

→ Oxidation states are valencies with +(-) before them i.e. $+n$ or $-n$.
Remember;

✓ Valency → Is the number of electrons
 lost/gained (shared)

Information obtained from **periodic table** (electronic configuration)



Groups $\begin{matrix} I \\ II \\ III \\ IV \\ V \\ VI \\ VII \\ VIII/0 \end{matrix}$

Metals (+) → Have positive oxidation states i.e. +1, +2, +3.

Non-Metals (-) → Have negative oxidation states i.e. -4, -3, -2, -1.

$VIII/0$ → unreactive → ~~or~~ 0

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TopNotch Note: When the compound has a sum NO overall charge

of * Neutral compounds have e.g. H_2SO_4

* Single elements also have an oxidation state of 0 e.g. Na^0 alone, Fe^0 alone, I^0 alone, etc

* So when do oxidation states work?

Answer: They work for elements ^{already} within compounds!

Na $\rightarrow$ alone. $\checkmark$

oxidation state = 0

$\rightarrow$ The whole $NaCl$ has an oxidation state of 0

(a). In what is the oxidation state of Na in $NaCl$?

Let the oxidation state of Na be x .

$NaCl$

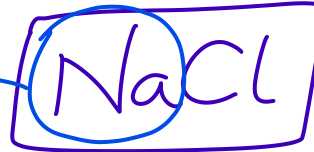
{ oxidation state of Na + oxidation state of Cl } = 0

$$x + (-1) = 0$$

$$x = +1$$

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⑥ what is the oxidation state of Cl in NaCl ?



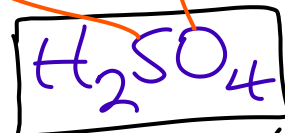
Let the oxidation state of Cl be y .

$$+1 + y = 0$$

$$y = -1$$

[2] Oxidation of

(i). Hydrogen (H) in H_2SO_4
Let the oxidation state of H be x



usually -2

usually -2

(ii). Sulphur (S) in H_2SO_4

(iii). Oxygen (O) in H_2SO_4

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NOTE: As we calculate oxidation states, have it mind that abnormalities occur;

(Hydrogen sulphide)

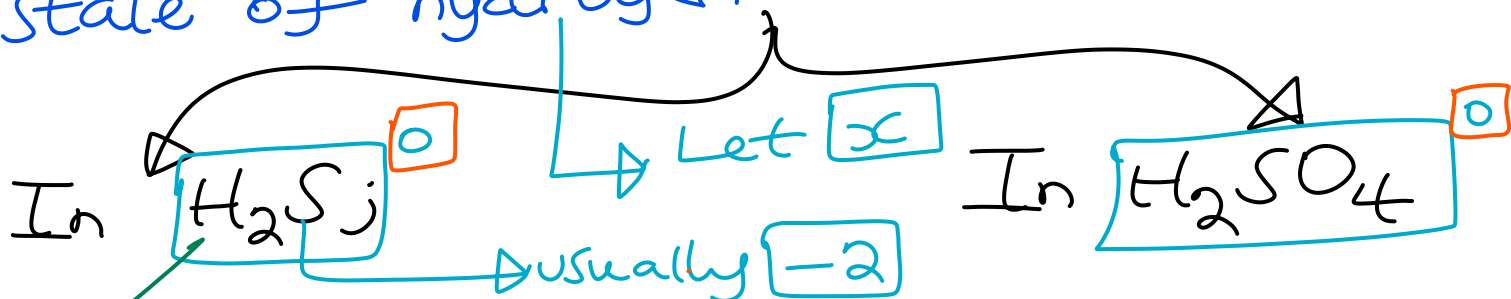


VS

(Sulphuric Acid)



Task: Calculate the oxidation state of hydrogen in each case!

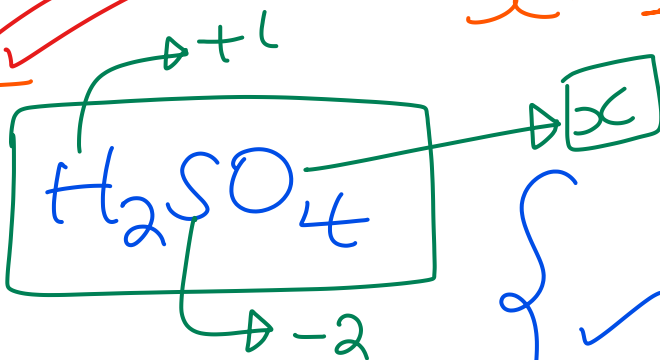


$$2x + (-2) = 0 \quad \text{VS} \quad 2x + (-2) + 4(-2) = 0$$

$$x = +1$$

$$x = +5$$

Qn: 0 in



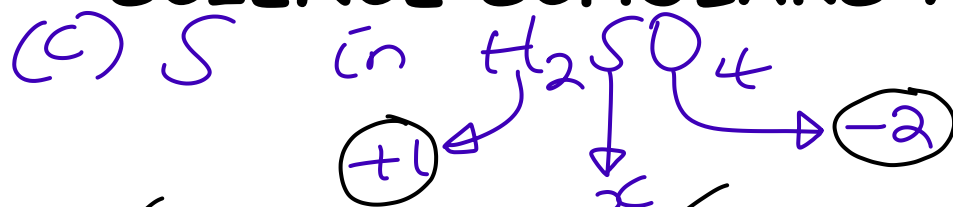
$$2(+1) + (-2) + 4x = 0$$

$$2 + -2 + 4x = 0$$

$$* \quad x = 0 \quad *$$

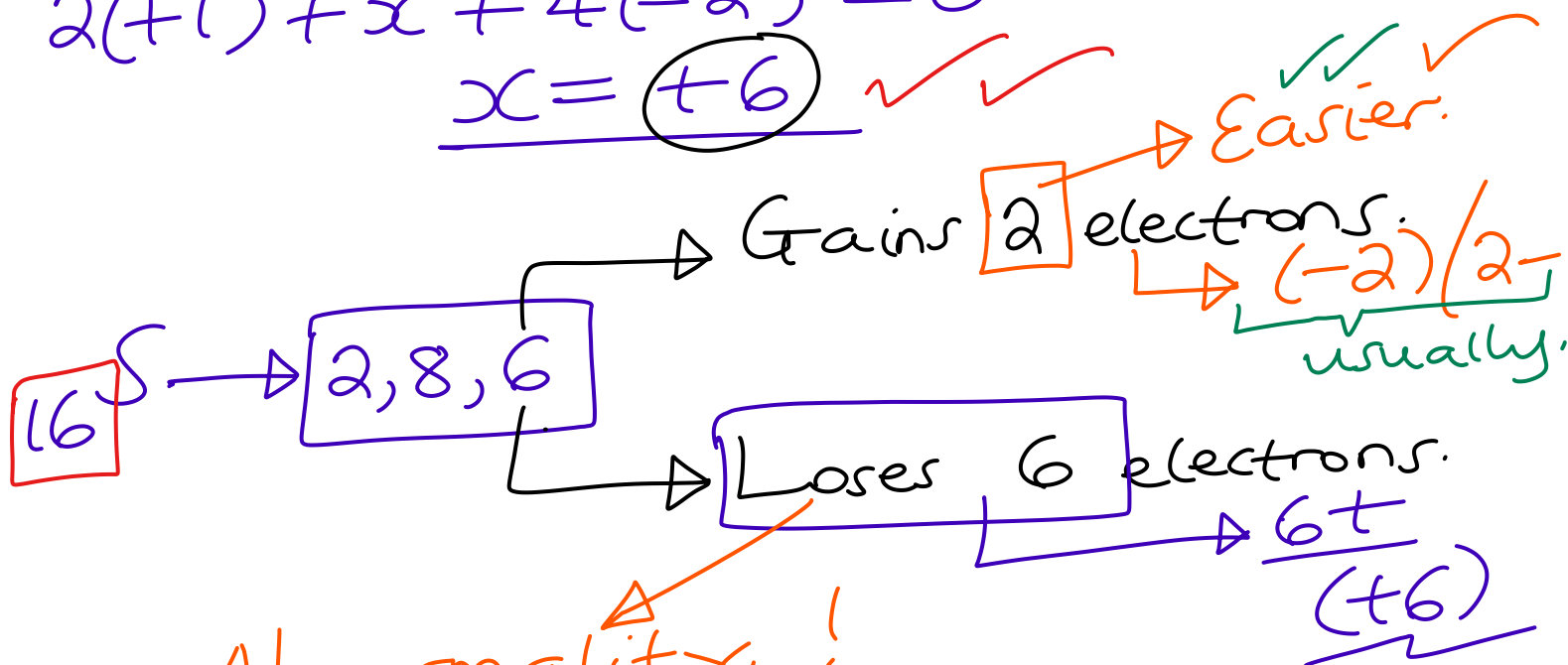


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$$2(+1) + x + 4(-2) = 0$$

$$x = +6$$



Those values are ONLY used to calculate.

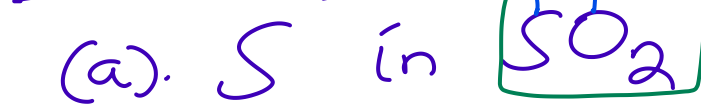
(+1) (+2) (+3) (+4) (-1) (-2) (-3) (-4) 0
 I II III IV V VI VII VIII

Group

BUT; the answer you get MAY/MAYNOT BE THE EXACT DEPENDING ON THE CPD $\rightarrow$ ✓

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Last one; ✓



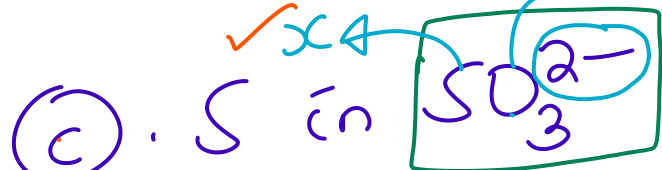
$$x + 2(-2) = 0$$

$$x = +4$$



$$x + 3(-2) = 0$$

$$x = +6$$



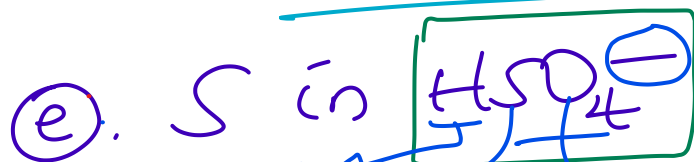
$$x + 3(-2) = -2$$

$$x = +4$$



$$(+1) + x + 3(-2) = -1$$

$$x = +4$$



$$(+1) + x + 4(-2) = -1$$

$$x = +6$$



$$y + 4(-2) = -2$$

$$y = +6$$



$$2(+1) + b = 0$$

$$b = -2$$

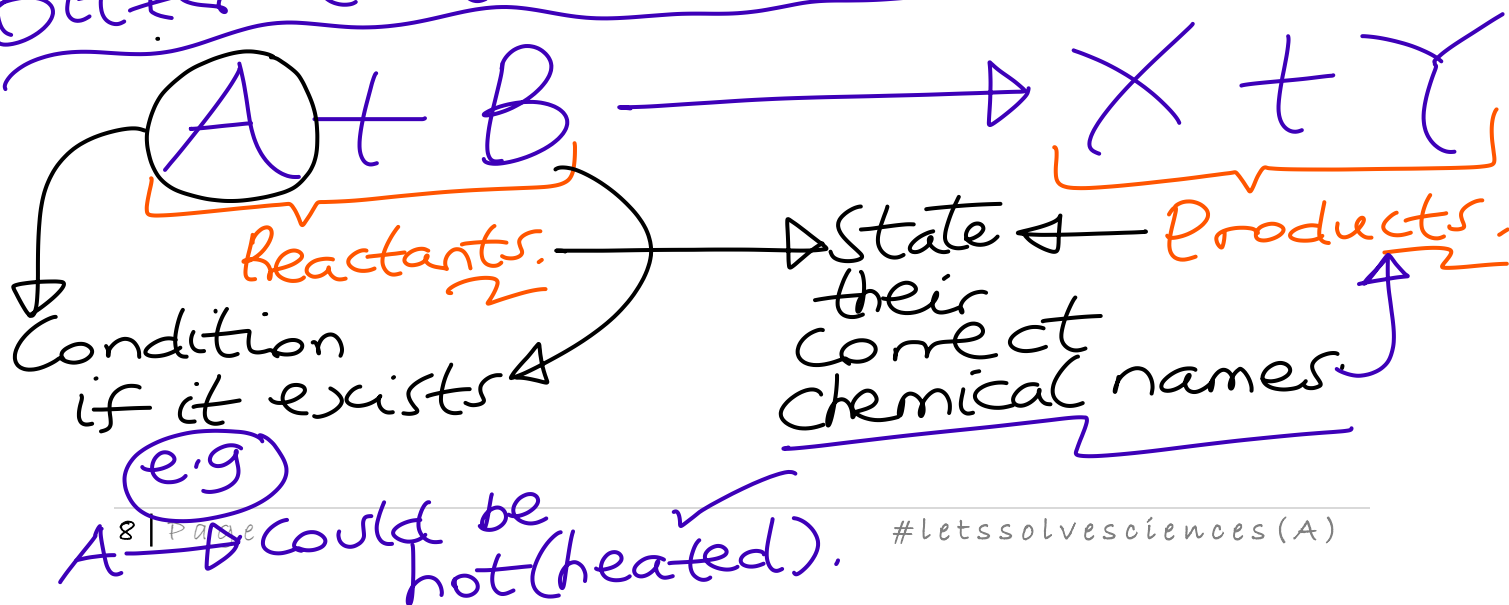
Note: USE THE PERIODIC TABLE TO CALCULATE BUT YOUR ANSWER MAY/MAYNOT RELATE TO THE PERIODIC TABLE.

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Brain teasers:

- (a). P in H_3PO_4 $\rightarrow +5$ ✓
 (b). P in H_2PO_2^- $\rightarrow +1$ ✓
 (c). P in PH_3 $\rightarrow -3$ ✓
 (d). P in H_3PO_3 $\rightarrow +3$ ✓
 (e). P in HPO_4^{2-} $\rightarrow +5$ ✓
 (f). P in HPO_3^{2-} $\rightarrow +3$ ✓
 (g). P in H_2PO_4^- $\rightarrow +5$ ✓
 (h). P in H_2PO_3^- $\rightarrow +3$ ✓
 (i) P in PCl_5 $\rightarrow +5$ ✓ ✓

Bitter truth about reactions:



Note: In Most CASES; for steam to
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react with something X, X must be
hot (heated (strongly
heated (red hot).

Anormally

↳ deviates from
the norm;

CONDITIONS: Can be at Room temperature {No need
of saying it.}

①. Elements

↳ Can be heated (burnt
(Please say it)



②. Water → Cold water
→ STEAM → SAY IT (Commonest)
→ hot water
→ warm water



③. Acids → dilute (cold, warm)
↓
conditions

→ concentrated (cold, hot)
(RESEARCH)



④. ALKALIS → Cold dilute
→ hot concentrated

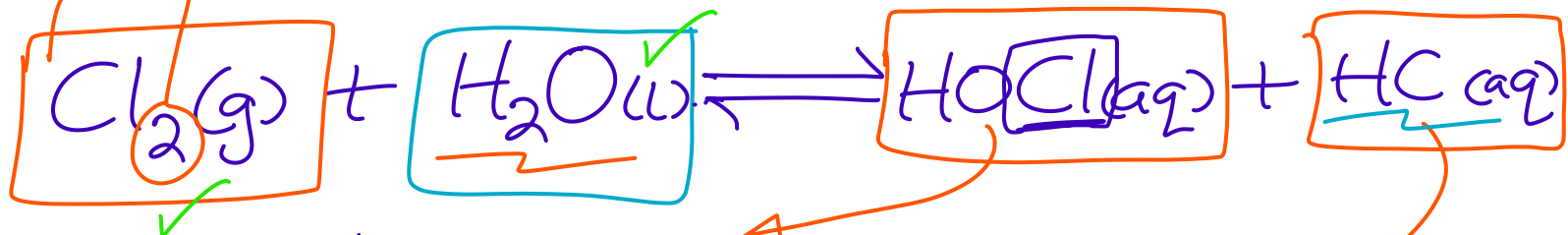
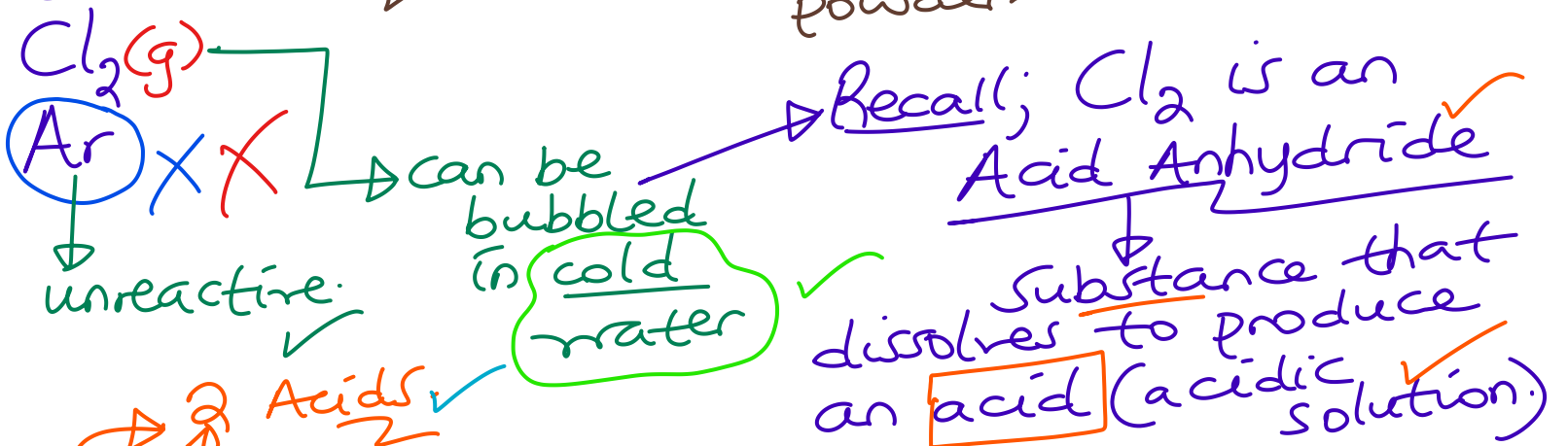
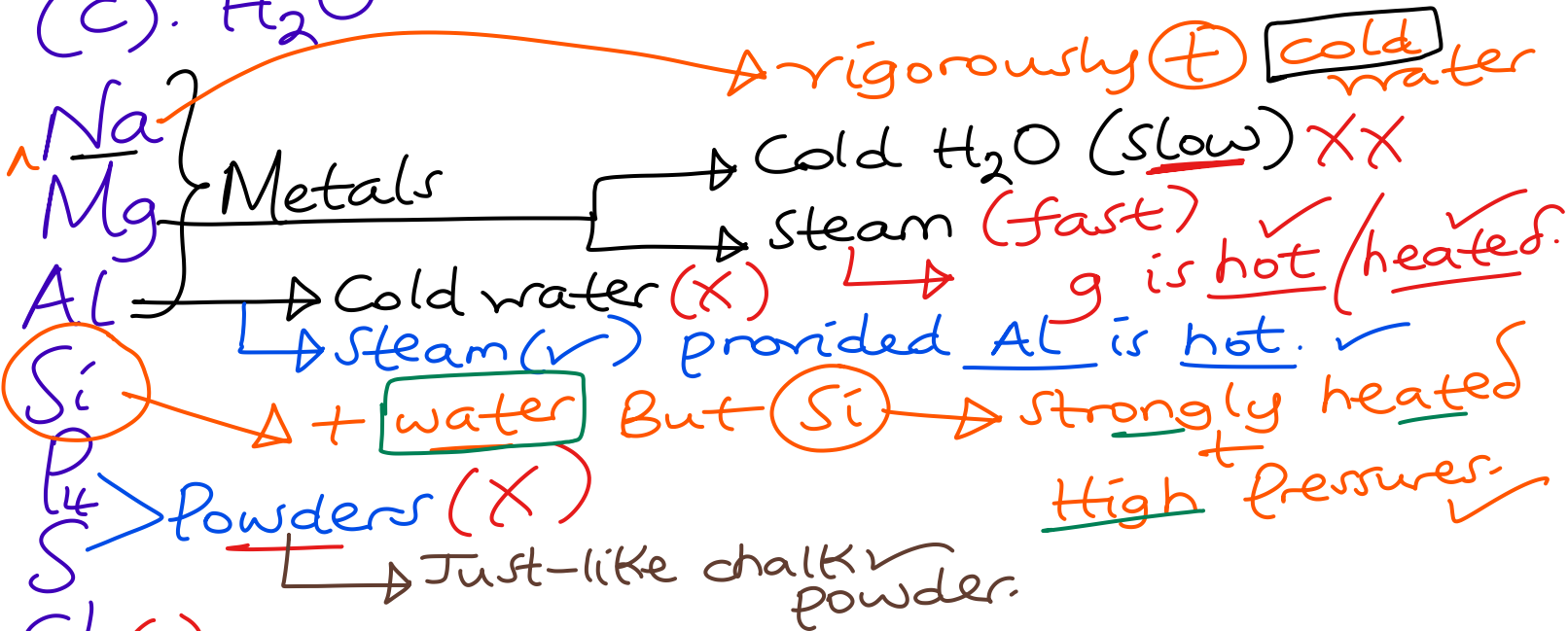


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* Statements describing the reaction are derived from a correct equation.

Reactions of elements with;

(c). H_2O



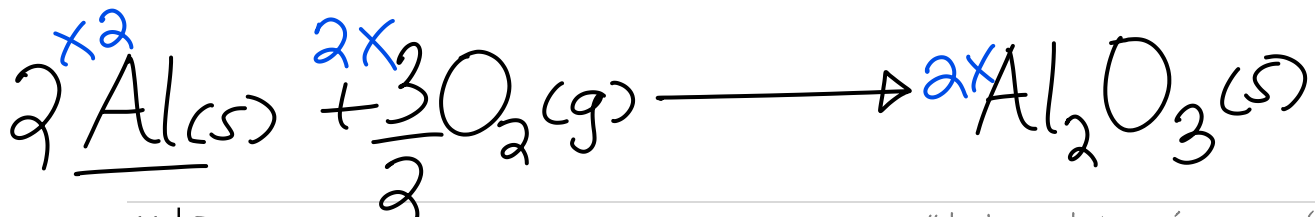
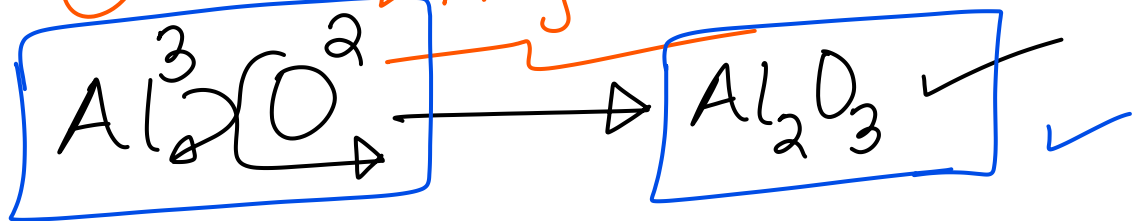
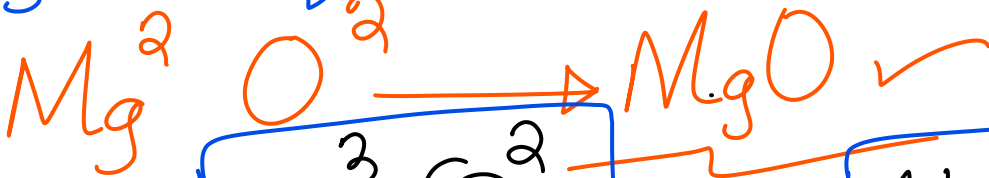
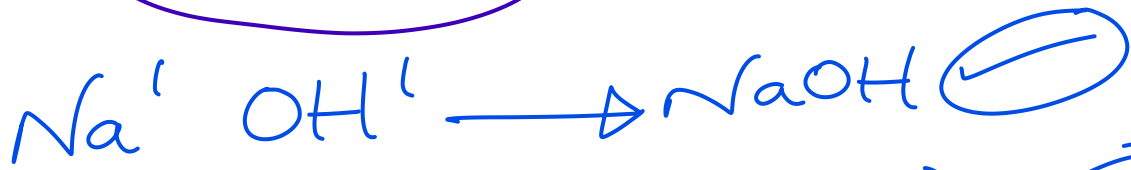
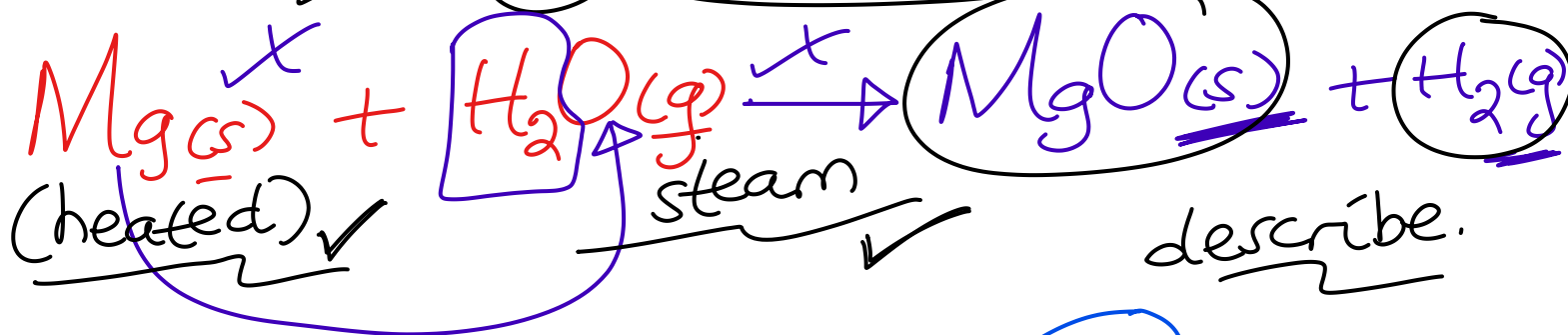
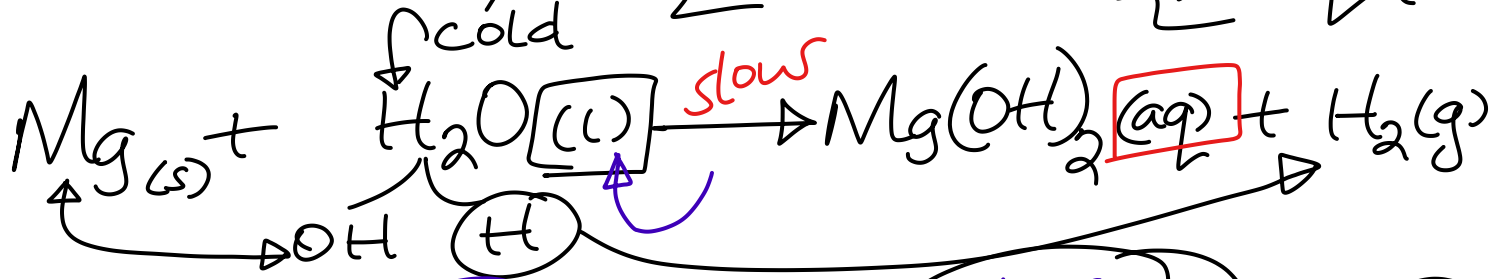
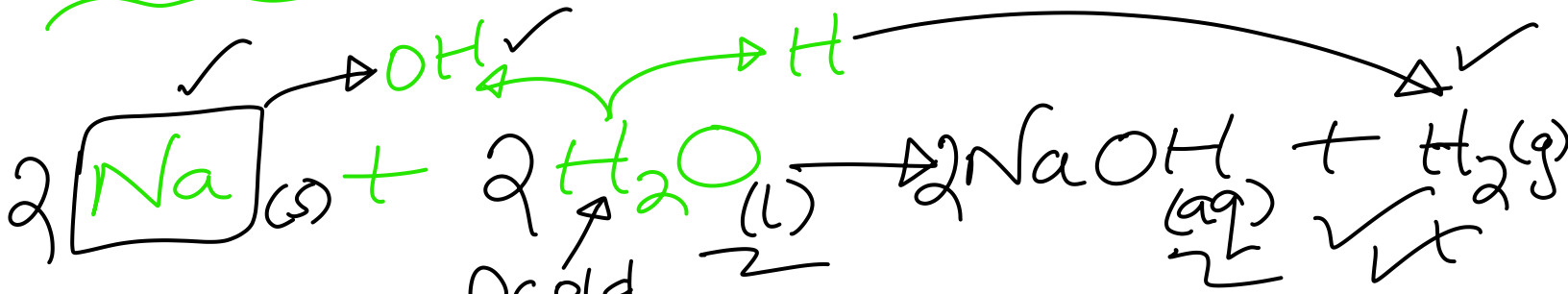
Chloric(I) Acid $\checkmark$ because $\checkmark$

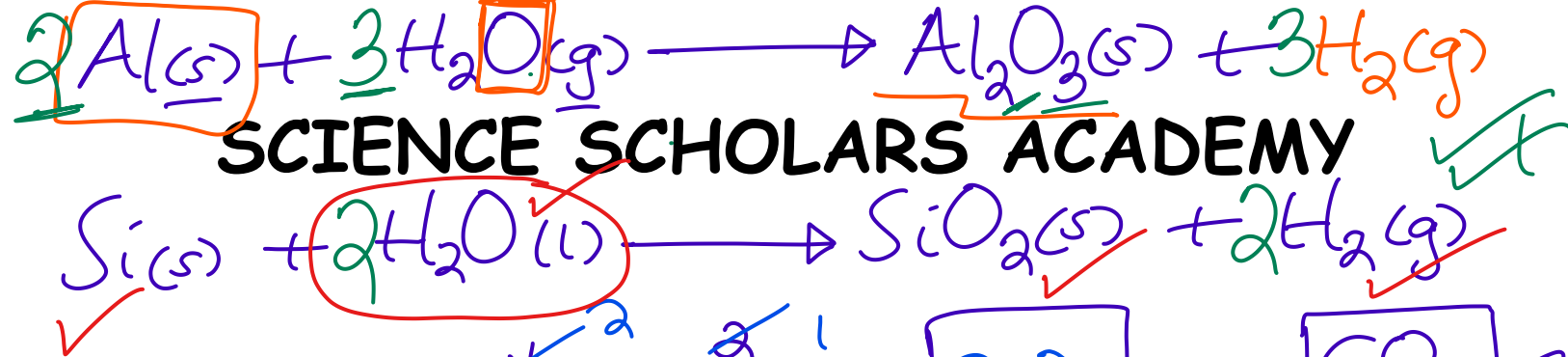
Hydro-chloric Acid $\checkmark$

Oxidation state of Cl in $\text{HOCl} = +1$ $\checkmark$

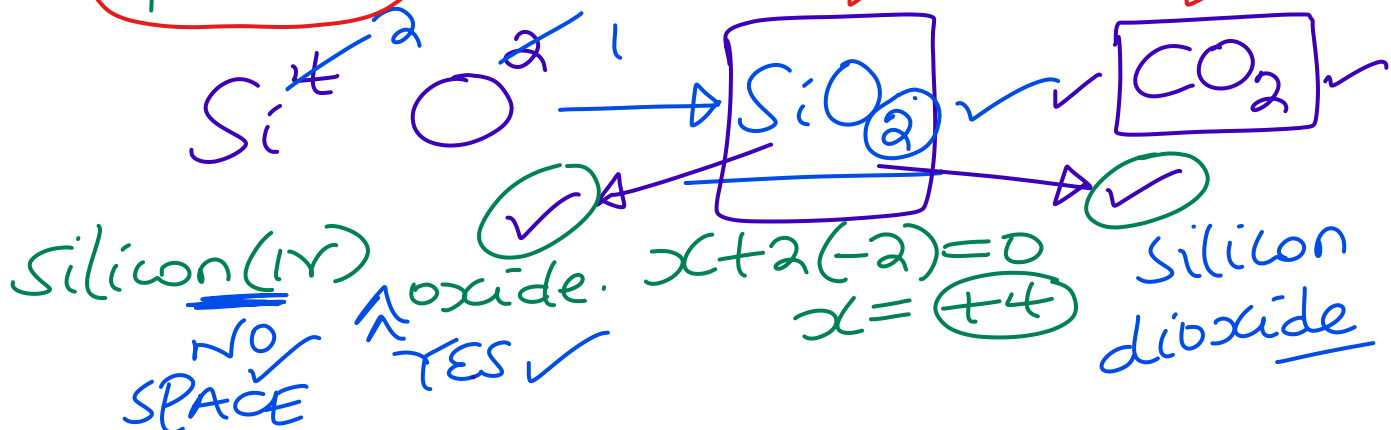
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Behind the scenes: ✓





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Note: (oxidation states)

-ve $\xrightarrow{\text{oxiditation}}$ +ve

+ve $\xrightarrow{\text{Reduction}}$ -ve

{Answer using English
not chemistry}

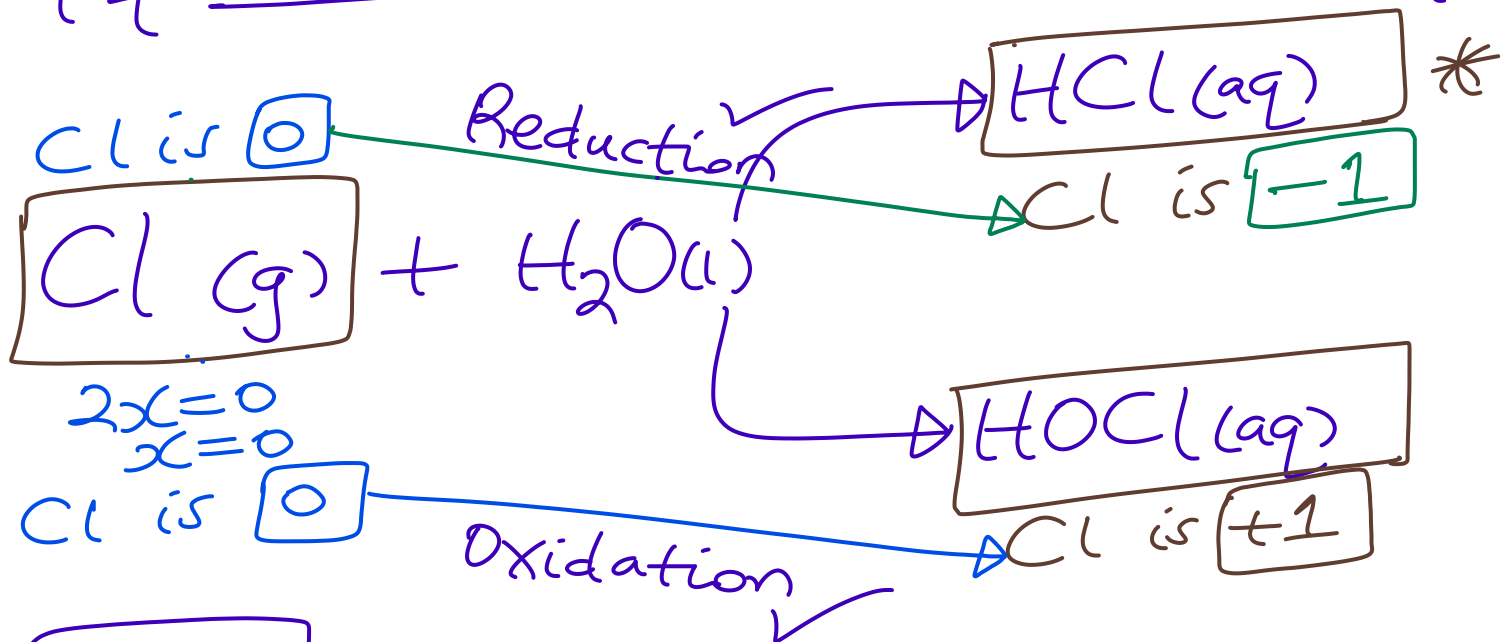
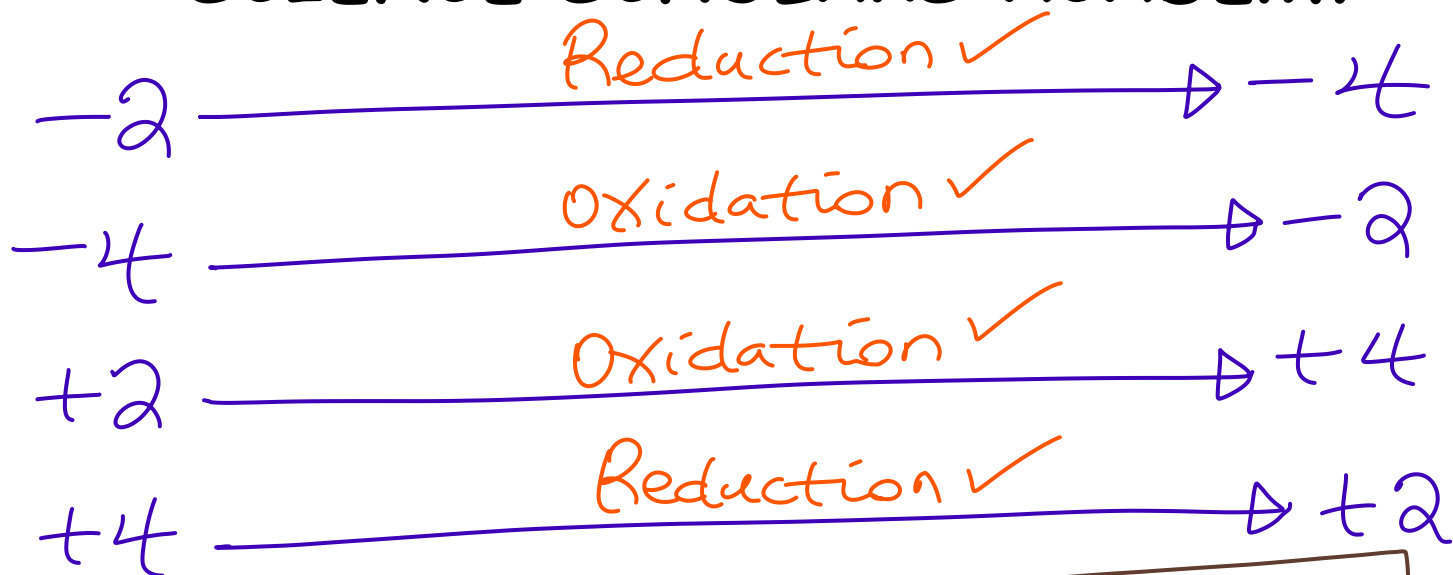
-ve $\xrightarrow{\text{Oxidation } \checkmark}$ 0

0 $\xrightarrow{\text{Reduction } \checkmark}$ -ve

+ve $\xrightarrow{\text{Reduction } \checkmark}$ 0

0 $\xrightarrow{\text{oxidation } \checkmark}$ +ve

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NOTE: A disproportionation reaction is one in which the **SAME SPECIES** undergoes **SIMULTANEOUS**

happen at the same time in the same **Eqn**

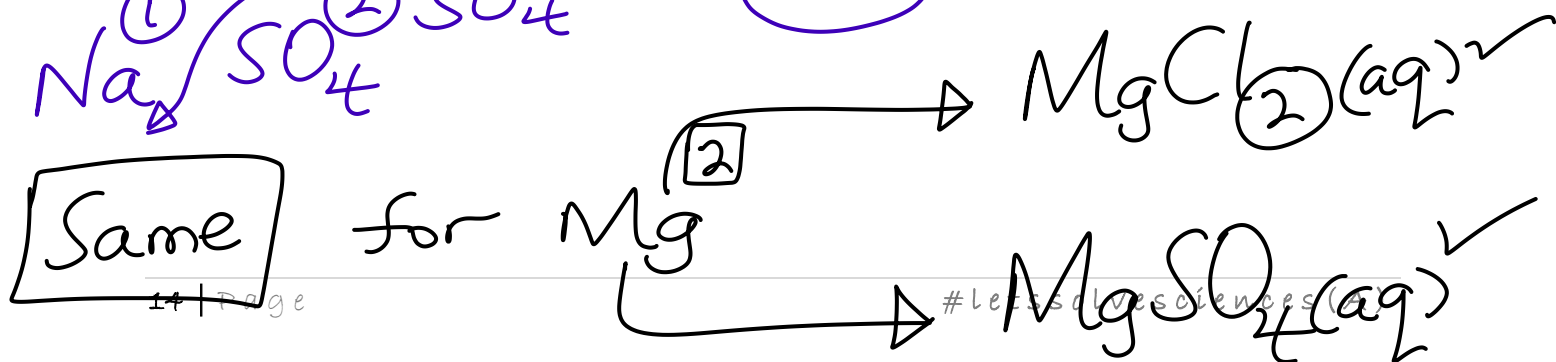
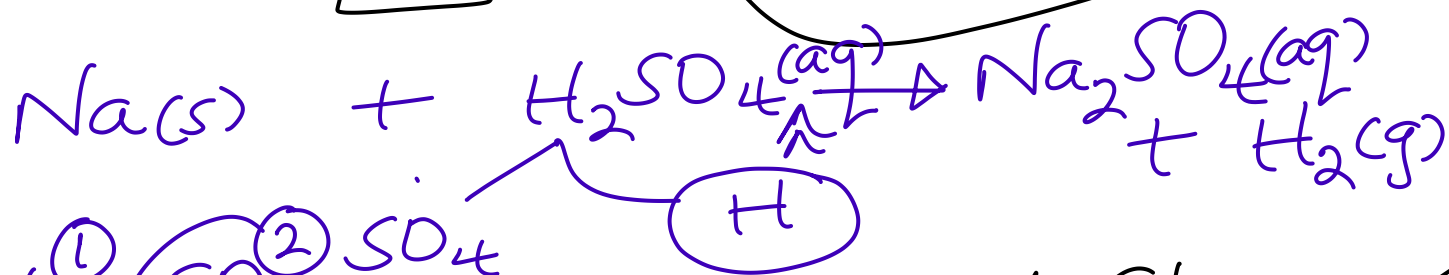
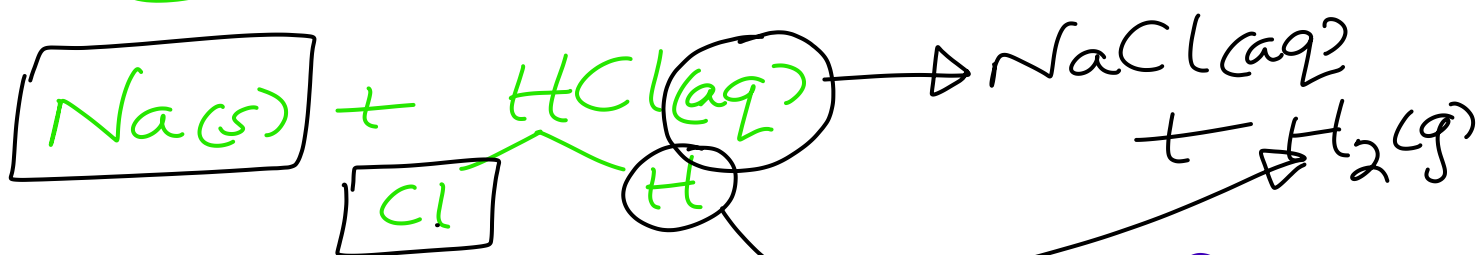
oxidation
 &
 reduction

So $\rightarrow \text{Cl}_2$ disproportionates in H_2O

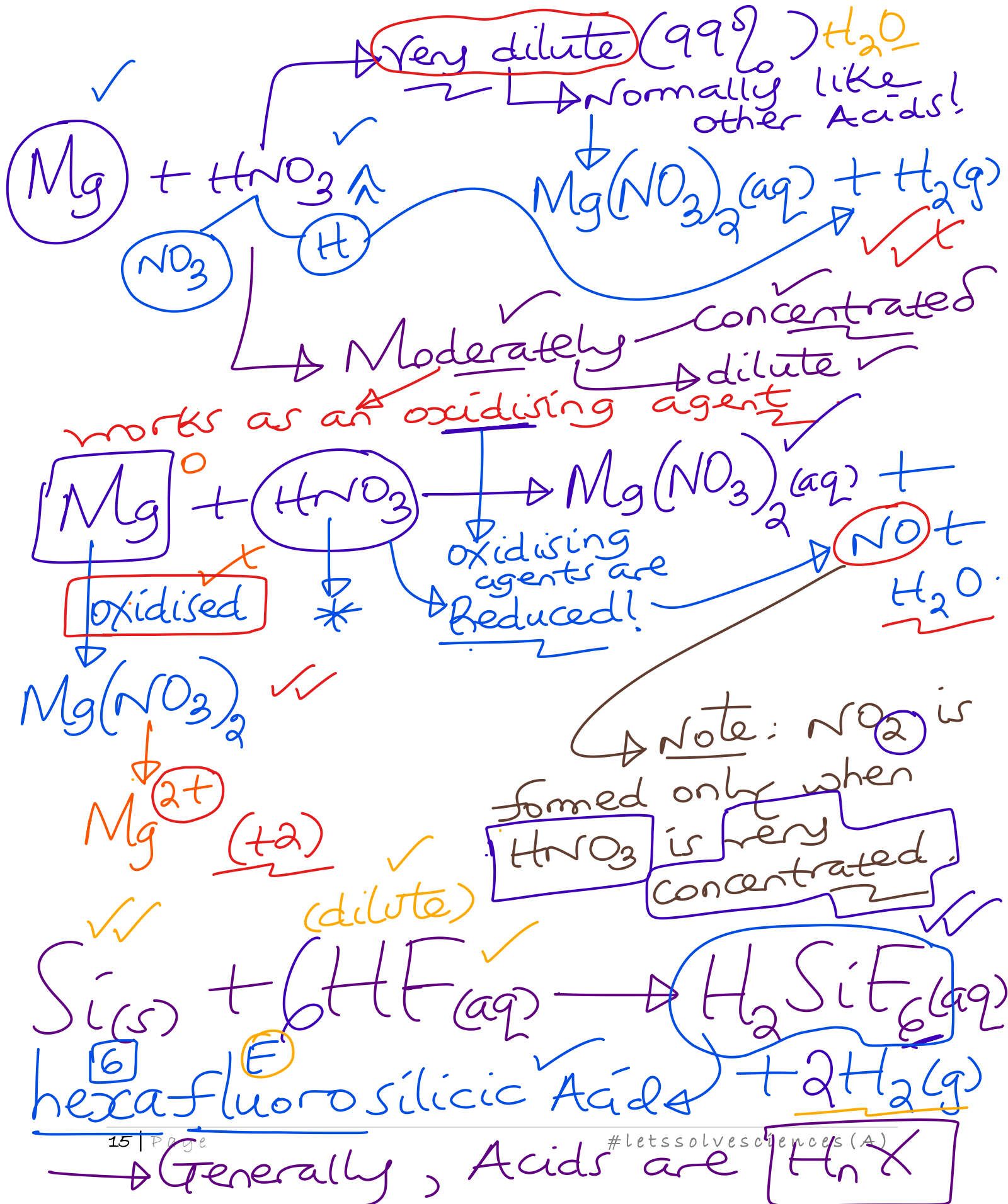
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(D). Dilute Acids

Na } takes over H^+ O_3 } very dilute.
 Sg } All react with HCl } Moderately dilute.
 Al } ONLY does not react with H_2SO_4 } dilute
 Si } ONLY Si reacts with HF } Hydrofluoric Acid!
 P_4 } XX because dilute Acids have a lot of H_2O & $\text{P}_4(\text{S})$ don't like H_2O .
 S }
 Cl_2 } Acid Anhydride (Acidic) } Can't react with (Acids)
 Ar } Behind the scenes



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RESEARCH:

(E). Concentrated Acids

- HCl
- H_2SO_4
- HNO_3

END

Imreh
Paul Tutor

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Ug

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CALLS. }

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